

4.3 Topic 7: Run for your life

Students will be assessed on their ability to:

- 1 Demonstrate knowledge and understanding of the *How Science Works* areas listed in the table on page 13 of this specification.
- 2 Describe the structure of a muscle fibre and explain the structural and physiological differences between fast and slow twitch muscle fibres.
- 3 Explain the contraction of skeletal muscle in terms of the sliding filament theory, including the role of actin, myosin, troponin, tropomyosin, calcium ions (Ca^{2+}), ATP and ATPase.

- 4 Recall the way in which muscles, tendons, the skeleton and ligaments interact to enable movement, including antagonistic muscle pairs, extensors and flexors.
- 5 Describe the overall reaction of aerobic respiration as splitting of the respiratory substrate (eg glucose) to release carbon dioxide as a waste product and reuniting of hydrogen with atmospheric oxygen with the release of a large amount of energy.
- 6 **Describe how to investigate rate of respiration practically.**
- 7 Recall how phosphorylation of ADP requires energy and how hydrolysis of ATP provides an accessible supply of energy for biological processes.
- 8 Describe the roles of glycolysis in aerobic and anaerobic respiration, including the phosphorylation of hexoses, the production of ATP, reduced coenzyme and pyruvate acid (details of intermediate stages and compounds are not required).
- 9 Describe the role of the Krebs cycle in the complete oxidation of glucose and formation of carbon dioxide (CO_2), ATP, reduced NAD and reduced FAD (names of other compounds are not required) and that respiration is a many-stepped process with each step controlled and catalysed by a specific intracellular enzyme.
- 10 Describe the synthesis of ATP by oxidative phosphorylation associated with the electron transport chain in mitochondria, including the role of chemiosmosis and ATPase.
- 11 Explain the fate of lactate after a period of anaerobic respiration in animals.
- 12 Understand that cardiac muscle is myogenic and describe the normal electrical activity of the heart, including the roles of the sinoatrial node (SAN), the atrioventricular node (AVN) and the bundle of His, and how the use of electrocardiograms (ECGs) can aid the diagnosis of cardiovascular disease (CVD) and other heart conditions.

- 13 Explain how variations in ventilation and cardiac output enable rapid delivery of oxygen to tissues and the removal of carbon dioxide from them, including how the heart rate and ventilation rate are controlled and the roles of the cardiovascular control centre and the ventilation centre.
- 14 **Describe how to investigate the effects of exercise on tidal volume and breathing rate using data from spirometer traces.**
- 15 Explain the principle of negative feedback in maintaining systems within narrow limits.
- 16 Discuss the concept of homeostasis and its importance in maintaining the body in a state of dynamic equilibrium during exercise, including the role of the hypothalamus and the mechanisms of thermoregulation.
- 17 Explain how genes can be switched on and off by DNA transcription factors including hormones.
- 18 Analyse and interpret data on possible disadvantages of exercising too much (wear and tear on joints, suppression of the immune system) and exercising too little (increased risk of obesity, coronary heart disease (CHD) and diabetes), recognising correlation and causal relationships.
- 19 Explain how medical technology, including the use of keyhole surgery and prostheses, is enabling those with injuries and disabilities to participate in sports, eg cruciate ligaments repair using keyhole surgery and knee joint replacement using prosthetics.
- 20 Outline two ethical positions relating to whether the use of performance-enhancing substances by athletes is acceptable.